BIAS SIMILARITY MEASUREMENT: A BLACK-BOX AUDIT OF FAIRNESS ACROSS LLMS

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ABSTRACT

Conventional fairness evaluations assess large language models (LLMs) in isolation, producing scalar scores that obscure how biases align, diverge, or persist across model families and release generations. We introduce **Bias Similarity Measurement (BSM)**, a black-box framework that reframes fairness as a *relational* property by comparing models' bias signatures across six complementary views: accuracy, directional bias scores, output histograms, cosine output distance, abstention (UNK) flip rates, and CKA-based representational similarity. BSM evaluates models on bias-sensitive prompts from BBQ Parrish et al. (2022), UnQover Li et al. (2020), and StereoSet Nadeem et al. (2021), and constructs pairwise similarity matrices within and across model families to reveal inheritance, convergence, and drift. We apply BSM to 30 LLMs spanning LLaMA, Gemma, GPT, and Gemini families over more than one million prompts—the largest bias similarity study to date. Our key finding is that instruction tuning primarily induces abstention rather than genuine representational debiasing: Gemma 2 9B-It and Gemma 3 12B-It flip over 50% of biased answers to "Unknown," while base-to-tuned CKA similarity remains above 0.94, indicating that internal representations change minimally. Furthermore, small models can *regress* under tuning: LLaMA 3.2 3B's disambiguated bias score worsens from 17.67 to 30.97 under forced-choice evaluation. BSM exposes when apparent fairness improvements reflect refusal policies rather than genuine debiasing, and demonstrates that open-weight models can match proprietary ones on relational fairness metrics.

1 INTRODUCTION

As large language models (LLMs) are increasingly deployed in high-stakes domains—hiring, healthcare, legal services, and beyond—auditing their fairness has become a societal imperative Ferrara (2023); Gallegos et al. (2024). Yet the dominant evaluation paradigm assesses each model in isolation, computing a scalar bias score and subsequently comparing scores across systems. This approach has two critical blind spots. First, it masks relational structure: models that share training lineage or data may exhibit *inherited* biases that per-model scores cannot reveal. Second, it conflates fundamentally different fairness mechanisms—a model can achieve a low bias score by refusing to answer (abstaining) rather than by encoding balanced representations. Distinguishing *fairness through caution* from *fairness through representation* is essential for procurement decisions, regression testing across model versions, and lineage auditing.

Prior work on LLM bias evaluation has focused primarily on curated benchmarks such as BBQ Parrish et al. (2022), StereoSet Nadeem et al. (2021), and UnQover Li et al. (2020), or on comprehensive surveys that catalog evaluation approaches without providing a unified cross-model comparison framework Gallegos et al. (2024); Kumar et al. (2024). Studies of instruction-tuning effects Gupta et al. (2024) have demonstrated that alignment procedures alter output distributions significantly, but the underlying mechanism—whether tuning reshapes internal representations or merely adds surface-level refusal behavior—is poorly understood. Meanwhile, model similarity research Kornblith et al. (2019); Klabunde et al. (2025) has developed powerful representation comparison tools that have not yet been systematically applied to the fairness domain.

We address these gaps by introducing **Bias Similarity Measurement (BSM)**, a unified black-box framework that measures fairness *relationally*. BSM constructs a multi-view bias signature for each model by computing six metrics over bias-sensitive prompts, then assembles pairwise similarity

matrices to map fairness inheritance, convergence, and drift across model families. A key design choice is the separation of *ambiguous* contexts (where abstention is appropriate) from *disambiguated* contexts (where an answer is required), which disentangles the two fairness mechanisms described above.

We apply BSM to 30 LLMs spanning LLaMA Touvron et al. (2023); Dubey et al. (2024); Meta AI (2024), Gemma Team et al. (2024a;b; 2025), GPT Radford et al. (2019); OpenAI (2024; 2025), and Gemini Google AI Developers (2025) families, evaluating over one million prompts. Our study yields four main contributions and findings:

1. **BSM as a relational fairness framework.** We introduce the first unified black-box framework that measures fairness as a comparative property via scalar, distributional, behavioral, and representational similarity views.
2. **Instruction tuning primarily induces abstention.** Tuned models flip large fractions of biased answers to "Unknown" (over 50% for Gemma 2 9B-It and Gemma 3 12B-It), while CKA scores confirm that internal representations change minimally (> 0.94 diagonal).
3. **Small-model regressions.** LLaMA 3.2 3B's bias score *worsens* under instruction tuning (s_{DIS} : $17.67 \rightarrow 30.97$), a failure mode invisible to isolated-model audits.
4. **Open-weight parity with proprietary models.** Gemma 3 12B-It and Gemma 3 27B-It achieve near-neutral bias scores competitive with GPT-4o Mini, demonstrating that the fairness gap between open and proprietary models is closing.

BSM supports practical workflows for procurement (relational comparison of candidate models), regression testing (detecting fairness drift across releases), and lineage screening (identifying bias inheritance from base to tuned models). Our central takeaway is that fairness should be audited as comparative bias similarity, not isolated scores—only then can we distinguish genuine debiasing from refusal policies.

2 Related Work

2.1 Bias Benchmarks and Evaluation

A rich ecosystem of benchmarks probes stereotyping and discriminatory behavior in language models. **BBQ** Parrish et al. (2022) provides hand-crafted multiple-choice items designed to elicit demographic biases across nine social categories; crucially, each item appears in both an *ambiguous* version (where "Unknown" is the correct answer) and a *disambiguated* version (where context resolves the question), enabling study of whether models abstain appropriately or answer stereotypically. **StereoSet** Nadeem et al. (2021) measures stereotypical bias in open-ended text completion by asking models to choose between stereotypical, anti-stereotypical, and neutral completions, yielding a stereotype score alongside a language model quality score. **CrowS-Pairs** Nangia et al. (2020) adopts a contrastive approach for masked language models, comparing stereotypical and anti-stereotypical sentence pairs. **UnQover** Li et al. (2020) frames bias measurement as a forced-choice task using underspecified questions: because no answer is justified by context, any systematic preference reveals stereotypical leanings.

Comprehensive surveys Gallegos et al. (2024); Kumar et al. (2024) catalog the diversity of bias types and evaluation methodologies. Recent work such as CEB Wang et al. (2025) and BEATS Abhishek et al. (2025) propose more structured evaluation protocols. Stureborg et al. (2024) highlight the inconsistency of LLM self-evaluations. Oketunji et al. (2023) propose the LLM Bias Index for tracking bias across model versions, and Zhao et al. (2023) introduce GPTBias as a comprehensive framework for GPT-family bias evaluation. Importantly, Kumar et al. (2025) confirm that no current LLM is free from measurable bias, underscoring the need for systematic cross-model auditing. BSM complements these efforts by adding a cross-model relational dimension absent from prior work.

2.2 Effects of Instruction Tuning on Fairness

Instruction tuning and RLHF are standard post-training procedures for modern LLMs Touvron et al. (2023); Dubey et al. (2024); Team et al. (2025), but their fairness implications are complex. Gupta et al. (2024) show that aligned models can exhibit reduced bias on benchmarks, but the reduction is

often mediated by increased refusal rates. Tjuatja et al. (2024) demonstrate that LLMs are sensitive to prompt framing in ways that inflate or deflate apparent fairness. Manvi et al. (2024) find that larger models tend to exhibit greater cultural biases, suggesting that scale alone does not guarantee fairness. BSM directly operationalizes the distinction between abstention-driven and representation-driven fairness via its UNK flip rate (M5) and CKA metrics (M6).

2.3 Model Similarity and Representational Comparison

Centered Kernel Alignment (CKA) Kornblith et al. (2019) provides a scale-invariant measure of representational similarity widely used to compare layers within and across neural networks. Abnar & Zuidema (2020) extend this framework to probing representational transfer. In the LLM context, Klabunde et al. (2025; 2023) apply representational similarity measures to 7B-parameter models and find substantial variation among architecturally similar systems. Li et al. (2021) propose ModelDiff for comparing model behaviors under distribution shift. Jin et al. (2024) apply similarity analysis for model fingerprinting and intellectual property protection. Liusie et al. (2024) and Bahrami et al. (2024) explore LLM comparison via distributional measures, which BSM also employs through Jensen–Shannon divergence Lin (1991) over output histograms. BSM adapts this model similarity paradigm specifically to the fairness domain, asking not whether models produce similar outputs in general, but whether they exhibit similar *patterns of bias*—a distinction with direct implications for understanding how bias propagates across model families. Bai et al. (2024) further demonstrate that implicit biases persist in models trained to be unbiased, suggesting a gap between output-level alignment and deeper representational change that BSM's CKA metric directly quantifies.

2.4 Practical Fairness Auditing

Beyond academic evaluation, there is growing demand for practical auditing tools supporting model procurement and governance Morehouse et al. (2025); Chaudhary et al. (2025); Fan et al. (2025). Dekoninck et al. (2025) propose comparative rating systems for LLMs. BSM addresses this need directly: its similarity matrices enable side-by-side comparison of model families, detection of regressions across versions, and screening for bias inheritance from base to tuned models.

3 Bias Similarity Measurement

Existing bias benchmarks typically report a single scalar score per model, collapsing a high-dimensional phenomenon into a point estimate that cannot capture *how* two models differ. We introduce **Bias Similarity Measurement (BSM)**, a black-box, relational framework that characterizes each model by a profile of six complementary metrics and then compares profiles pairwise to produce similarity matrices across an entire model population. Because BSM requires only input–output access, it applies equally to open-weight and proprietary systems.

Figure 1 illustrates the end-to-end pipeline. Given a set of N models and a bias benchmark, BSM proceeds in three stages: (i) prompt each model with every benchmark item and record its output; (ii) compute the six per-model metrics described below; and (iii) construct $N \times N$ similarity matrices—one per metric—using cosine similarity or Jensen–Shannon (JS) divergence Lin (1991) as appropriate.

Context separation. Following the BBQ protocol Parrish et al. (2022), every question appears in two forms: an *ambiguous* context that provides insufficient evidence to determine the correct answer, and a *disambiguated* context that resolves the question unambiguously. All six metrics are computed separately for each context type, yielding distinct views of a model's behavior under uncertainty versus under full information.

Metric 1: Accuracy. On disambiguated items the ground-truth answer is known, so we report standard accuracy. On ambiguous items the only defensible answer is "Unknown" (UNK); accuracy here measures the rate at which a model correctly abstains rather than guessing.

Metric 2: Bias scores (s_{AMB}, s_{DIS}). We adopt the signed bias score from BBQ Parrish et al. (2022), which ranges from -100 (maximally anti-stereotypical) through 0 (neutral) to $+100$ (max-

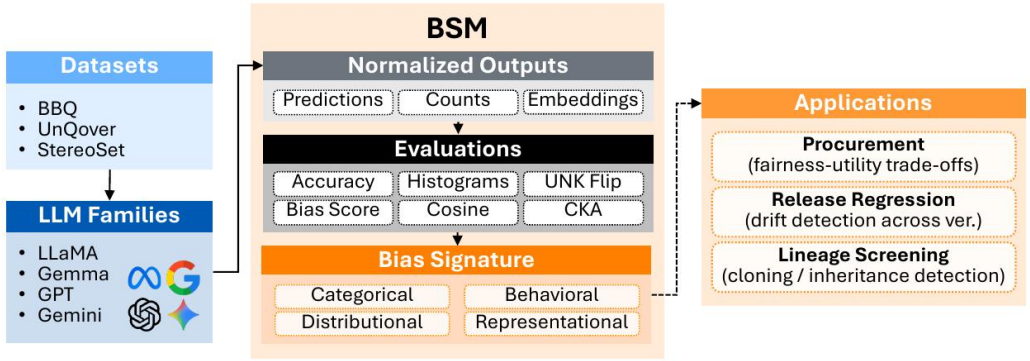

Figure 1: **BSM Pipeline.** Models are prompted with bias benchmarks, six complementary metrics are extracted, and pairwise similarity matrices are constructed to enable relational fairness comparisons.

imally stereotypical). The score is computed separately for ambiguous contexts (s_{AMB}) and disambiguated contexts (s_{DIS}), capturing whether errors are systematically aligned with social stereotypes in each setting.

Metric 3: Output histograms and JS divergence. For each demographic dimension (e.g., gender, race), we bin model outputs into a discrete histogram over answer choices. Pairwise JS divergence Lin (1991) between histograms quantifies distributional differences that scalar scores can obscure—for instance, two models with similar s_{AMB} may exhibit very different output distributions across demographic categories.

Metric 4: Cosine output similarity. We represent each model's output profile as a vector of per-category answer proportions and compute pairwise cosine similarity. This captures global alignment of output patterns across the full benchmark, complementing the category-level view provided by histograms.

Metric 5: UNK flip rate. For every ambiguous item we compare the base and instruction-tuned variant of each model. The *UNK flip rate* is the fraction of items on which the tuned model switches its answer to "Unknown" when the base model selected a demographic group. A high flip rate indicates that instruction tuning encourages abstention on under-specified questions, a desirable but potentially superficial debiasing strategy.

Metric 6: CKA representational similarity. Centered Kernel Alignment (CKA) Kornblith et al. (2019) measures the similarity of internal activation patterns between two models, providing a representation-level complement to the behavioral metrics above. Following Abnar & Zuidema (2020), we extract hidden-state representations from the last transformer layer for each benchmark prompt and compute linear CKA between base and tuned variants. We report both *diagonal* CKA (base vs. tuned within the same family) and *full* CKA (all pairs). High diagonal CKA paired with large behavioral shifts would indicate that instruction tuning changes *what* the model says without substantially altering *how* it represents the input.

Similarity matrices. Each metric yields an $N \times N$ matrix whose (i, j) entry quantifies how similarly models i and j behave under that lens. Taken together, the six matrices provide a multi-faceted relational map of fairness across the model population, enabling practitioners to identify clusters of similar bias behavior, detect regressions across model versions, and compare open-weight systems against proprietary baselines.

4 EVALUATION SETUP

Models. We evaluate 30 LLMs spanning four major families. The **LLaMA** lineage includes Vicuna, Alpaca, LLaMA 2 7B, LLaMA 3 8B/70B Dubey et al. (2024), LLaMA 3.1 8B/70B, and LLaMA 3.2 3B, each paired with its instruction-tuned variant where available. The **Gemma** family comprises Gemma 1 7B, Gemma 2 9B/27B, and Gemma 3 4B/12B/27B Team et al. (2025) with corresponding instruction-tuned models. We also include **Gemini** 1.5 Flash and 2.0 Flash as closed API-only models, and **GPT-2**, **GPT-4o Mini**, and **GPT-5 Mini** from OpenAI. This selection covers parameter counts from 3B to 70B, three generations of open-weight development, and both open and proprietary access regimes.

Datasets. Three established bias benchmarks are used. **BBQ** Parrish et al. (2022) provides multiple-choice questions in ambiguous and disambiguated contexts across nine demographic dimensions (age, disability, gender, nationality, physical appearance, race/ethnicity, religion, sexual orientation, socioeconomic status), with "Unknown" as an explicit answer option. **UnQover** Li et al. (2020) presents forced-choice stereotyping questions where the model must select between demographic groups, removing the abstention option. **StereoSet** Nadeem et al. (2021) elicits open-ended completions that are classified as stereotypical, anti-stereotypical, or neutral. In total, the evaluation encompasses over one million prompts.

Abstention handling. On BBQ, we treat selection of the "Unknown" option as abstention. For UnQover, where no UNK option exists, models that nevertheless output an abstention-like response (e.g., "I cannot answer") are recorded as UNK and excluded from the bias-score numerator, following the protocol of Li et al. (2020). This distinction is critical: it allows us to separate genuine debiasing from strategic refusal.

5 RESULTS

5.1 BIAS SCORE ANALYSIS

Table 1 reports average bias scores for all model families. The most striking result is the effect of instruction tuning on **LLaMA 3.1 8B**: s_{AMB} drops from 18.59 to 1.38 and s_{DIS} from 31.37 to 4.78, representing reductions of 93% and 85%, respectively. The larger LLaMA 3.1 70B is already near-neutral in its base form ($s_{\text{AMB}} = 0.42$, $s_{\text{DIS}} = 0.81$) and remains so after tuning.

In sharp contrast, **LLaMA 3.2 3B** exhibits a *regression*: instruction tuning increases s_{AMB} from 11.95 to 15.71 and s_{DIS} from 17.67 to 30.97. This is the only model in our study where tuning consistently worsens both bias scores, suggesting that alignment procedures effective at larger scales do not transfer straightforwardly to smaller architectures.

Table 1: **Average bias scores.** "–": anti-stereotypical, "+": stereotypical, and values near 0 = neutrality. Shown for ambiguous (s_AMB) and disambiguated (s_DIS) contexts.

Base Model	Avg. s_AMB		Avg. s_DIS	
	Base	Tuned	Base	Tuned
LLaMA 2 7B	5.45	4.30	7.50	6.65
LLaMA 3 8B	-4.78	-0.66	-8.72	-2.10
LLaMA 3 70B	-1.42	0.55	-4.51	2.50
LLaMA 3.1 8B	18.59	1.38	31.37	4.78
LLaMA 3.1 70B	0.42	-0.15	0.81	-1.44
LLaMA 3.2 3B	11.95	15.71	17.67	30.97
Gemma 7B	1.81	2.05	0.69	0.95
Gemma 2 9B	0.08	0.18	6.83	-2.02
Gemma 2 27B	6.95	0.51	14.31	-1.45
Gemma 3 4B	-3.89	5.83	2.69	8.62
Gemma 3 12B	4.36	0.15	6.12	-0.17
Gemma 3 27B	-1.25	0.07	-0.26	-1.50
Gemini 1.5	–	2.37	–	3.07
Gemini 2.0	–	-4.17	–	-5.54
GPT-2	72.43	–	96.19	–
GPT-4o Mini	–	0.47	–	2.66
GPT-5 Mini	–	0.21	–	1.10

Among proprietary models, **GPT-4o Mini** achieves near-neutral scores ($s_{\text{AMB}} = 0.47$, $s_{\text{DIS}} = 2.66$), and **GPT-5 Mini** approaches perfect neutrality ($s_{\text{AMB}} = 0.21$, $s_{\text{DIS}} = 1.10$). At the other extreme, **GPT-2** serves as a legacy baseline with extreme stereotypical bias ($s_{\text{AMB}} = 72.43$, $s_{\text{DIS}} = 96.19$), underscoring the progress made by modern alignment techniques.

Within the Gemma family, Gemma 2 27B and Gemma 3 12B/27B converge toward neutrality after tuning, with the tuned Gemma 3 27B reaching $s_{\text{AMB}} = 0.07$ and $s_{\text{DIS}} = -1.50$. Notably, Gemma 3 4B shows a pattern resembling LLaMA 3.2 3B: tuning shifts s_{AMB} from -3.89 to 5.83, flipping the bias direction from anti-stereotypical to stereotypical.

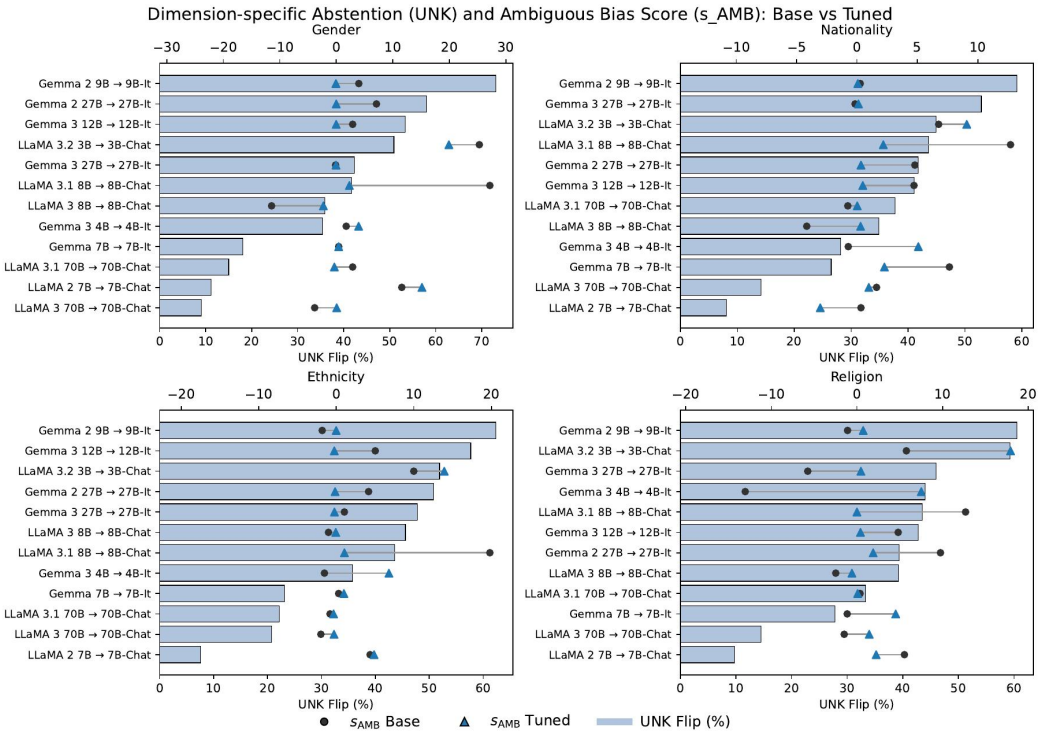

Figure 2: **Abstention and bias-flip rates.** Instruction tuning causes large increases in UNK selection for several models, with Gemma 2 9B-It and Gemma 3 12B-It flipping over 50% of biased answers to "Unknown."

5.2 Abstention and UNK Flip Rates

Figure 2 visualizes the abstention dynamics across model families. **Gemma 2 9B-It** exhibits the highest UNK flip rate, converting more than 50% of its base model's biased responses to "Unknown" after instruction tuning. **Gemma 3 12B-It** shows a similar pattern, also exceeding the 50% threshold. **LLaMA 3.1 8B-Instruct** flips approximately 40% of answers to UNK, consistent with its large bias-score improvement reported in Section 5.1.

These high flip rates raise an important interpretive question: does increased abstention reflect genuine debiasing—where the model has learned that the question is under-specified—or a surface-level strategy of refusing to answer whenever demographic content is detected? The distinction matters because a model that reflexively abstains will appear fair on BBQ-style benchmarks but may still harbor biased internal representations that manifest in open-ended settings. We return to this point in Section 6.

5.3 Representational Similarity

Table 2 reports CKA scores between base and instruction-tuned variants. Diagonal CKA values are uniformly high, ranging from 0.941 (Gemma 2 9B) to 0.991 (LLaMA 2 7B), indicating that instruction tuning preserves the vast majority of internal representational structure. Full-matrix CKA is lower but still above 0.85 for all pairs, confirming substantial cross-family representational overlap.

Table 2: Average CKA scores.

Model	Diag	Full
LLaMA 2 7B	0.991	0.902
LLaMA 3 8B	0.973	0.851
Gemma 1 7B	0.981	0.896
Gemma 2 9B	0.941	0.906
Gemma 3 12B	0.972	0.911

The combination of high CKA with large behavioral changes (e.g., LLaMA 3.1 8B's 93% bias-score reduction) suggests that instruction tuning operates primarily at the *decision layer*—altering which outputs are selected given

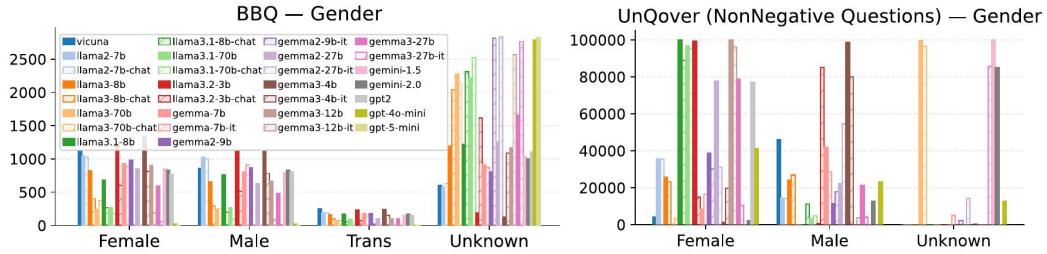

Figure 3: **Output distributions in the gender dimension.** Histograms reveal distributional differences that scalar bias scores can obscure, showing how models allocate probability mass across demographic answer choices.

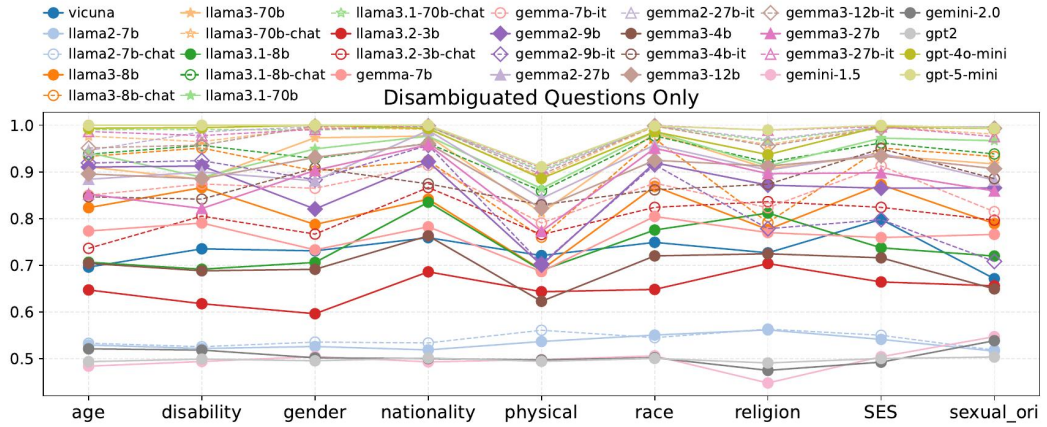

Figure 4: **Accuracy of LLMs on disambiguated BBQ questions.** Larger and more recent models generally achieve higher accuracy, with GPT-5 Mini leading across all categories.

a representation—rather than fundamentally restructuring how the model encodes inputs. Gemma 2 9B shows the lowest diagonal CKA (0.941), consistent with its aggressive abstention behavior: even representation-level changes are more pronounced when the model's output policy changes dramatically.

5.4 Output Distributions and Accuracy

Figure 3 shows output distributions for the gender dimension. Models with similar scalar bias scores can exhibit markedly different distributional profiles: for instance, two models near $s_{\text{AMB}} = 0$ may achieve neutrality through uniform randomness versus consistent UNK selection, patterns that are indistinguishable under scalar metrics but clearly differentiated by JS divergence over histograms.

Figure 4 reports accuracy on disambiguated BBQ questions. GPT-5 Mini achieves the highest accuracy across all demographic categories, followed by GPT-4o Mini and LLaMA 3.1 70B-Instruct. Accuracy generally increases with model scale and recency: LLaMA 3.x models substantially outperform LLaMA 2, and Gemma 3 models improve over Gemma 2. GPT-2 and early fine-tunes (Vicuna, Alpaca) show the lowest accuracy, consistent with their high bias scores.

5.5 Qualitative Model Summary

Table 3 synthesizes findings across all metrics. Several patterns emerge. First, *scale matters*: within each family, larger models consistently show lower bias and higher accuracy after tuning (e.g., LLaMA 3.1 70B vs. 8B, Gemma 3 27B vs. 4B). Second, *tuning is not uniformly beneficial*: LLaMA 3.2 3B and Gemma 3 4B demonstrate that alignment procedures can amplify bias in smaller models. Third, *open-weight models can match proprietary ones*: tuned Gemma 3 12B/27B achieve

Table 3: **Overall evaluation summary by model.** Qualitative synthesis of accuracy, abstention, bias direction, and representational similarity trends.

Model (abbrev.)	Key observations
Vicuna / Alpaca	Strongly anti-stereotypical; low accuracy; low abstention.
LLaMA 2 7B	Remain stereotypical after tuning; very low accuracy; weaker fairness.
LLaMA 3 8B	Anti-stereotypical lean; accuracy improves with tuning; moderate drift from LLaMA 2.
LLaMA 3.1 8B	Large bias drop after tuning; accuracy improves with tuning; high abstention.
LLaMA 3.1 70B	Near-neutral after tuning; high accuracy; high abstention even under forced choice.
LLaMA 3.2 3B	Strongly stereotypical; low accuracy; low abstention; weaker fairness vs 3.1 peers.
Gemma 2 9B	Stereotypical in base; abstention increases with tuning.
Gemma 3 4B	Slight stereotypical bias; accuracy competitive with LLaMA mid-size.
Gemma 3 12B / 27B	Near-neutral after tuning; high CKA similarity; fairness competitive with closed ones.
Gemini 1.5 / 2.0	Strong abstention; 2.0 skews anti-stereotypical; very low accuracy.
GPT-2	Extremely stereotypical bias; very low accuracy and fairness; serves as legacy baseline.
GPT-4o Mini	Near-zero bias; high accuracy; balanced abstention–fairness.
GPT-5 Mini	Near-perfect neutrality; highest accuracy; strongest stability across metrics.

bias scores competitive with GPT-4o Mini, and LLaMA 3.1 70B-Instruct approaches GPT-5 Mini on several metrics.

6 Discussion and Conclusion

Fairness as a relational property. BSM's central contribution is reframing bias evaluation from an absolute to a *relational* exercise. Rather than asking "is this model fair?" in isolation, BSM asks "how does this model's bias profile compare to others?" The six similarity matrices produced by BSM enable practitioners to identify clusters of models with similar bias behavior, detect regressions across versions, and benchmark open-weight systems against proprietary baselines—all without requiring access to model internals.

Abstention is not debiasing. Our results highlight a recurring pattern: instruction tuning dramatically increases UNK selection rates (exceeding 50% for Gemma 2 9B-It and Gemma 3 12B-It) while CKA scores remain above 0.94. This combination indicates that tuning teaches models to *refuse* rather than to *reason differently* about under-specified questions. While abstention is preferable to stereotypical guessing, it does not constitute genuine debiasing at the representational level and may not generalize to open-ended generation tasks where refusal is not a natural response.

Small-model regressions. The worsening of bias scores in LLaMA 3.2 3B (s_{DIS}: $17.67 \rightarrow 30.97$) and the direction flip in Gemma 3 4B (s_{AMB}: $-3.89 \rightarrow 5.83$) demonstrate that alignment techniques effective at 8B+ parameters do not transfer reliably to smaller architectures. This finding has practical implications: as smaller models are increasingly deployed on edge devices and in resource-constrained settings, their fairness properties require independent validation rather than extrapolation from larger siblings.

Open vs. proprietary models. GPT-5 Mini sets the current benchmark for neutrality ($s_{\text{AMB}} = 0.21$, $s_{\text{DIS}} = 1.10$), but open-weight alternatives are closing the gap. Tuned Gemma 3 27B ($s_{\text{AMB}} = 0.07$) and LLaMA 3.1 70B-Instruct ($s_{\text{AMB}} = -0.15$) achieve comparable bias scores, offering the additional advantage of reproducibility and auditability.

Practical applications. BSM's relational framework supports several deployment scenarios: (i) *model procurement*, where organizations can compare candidate models on fairness dimensions relevant to their use case; (ii) *regression testing*, where new model versions are evaluated against their predecessors to detect fairness degradation; and (iii) *lineage screening*, where similarity matrices reveal which model families share bias patterns, informing decisions about ensemble diversity.

Limitations and future work. Our study has several limitations. First, BSM currently relies on English-language benchmarks and may not capture bias patterns in multilingual settings. Second, CKA is computed only for open-weight models, leaving representational analysis incomplete for proprietary systems. Third, the six metrics, while complementary, do not exhaustively cover all dimensions of fairness. Future work should extend BSM to multilingual benchmarks, incorporate

additional fairness dimensions such as intersectional bias, and develop automated methods for interpreting the similarity matrices produced by the framework.

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